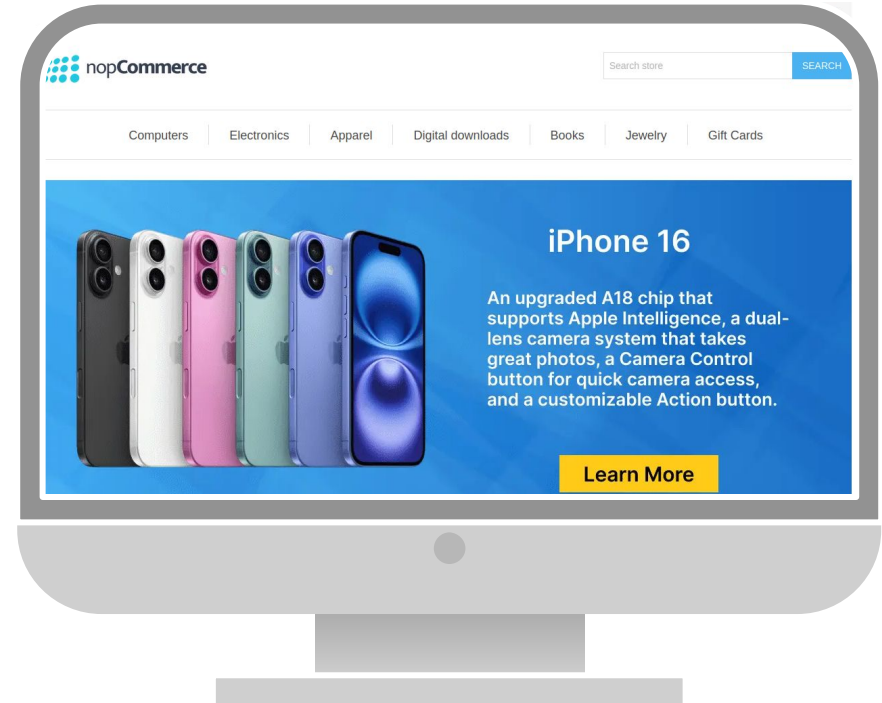
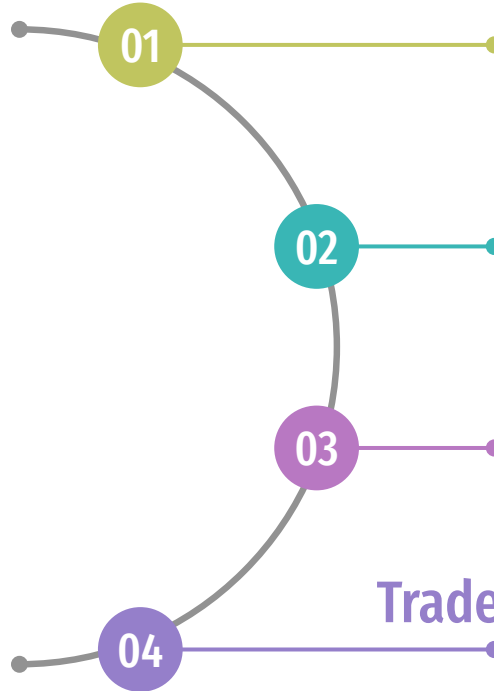
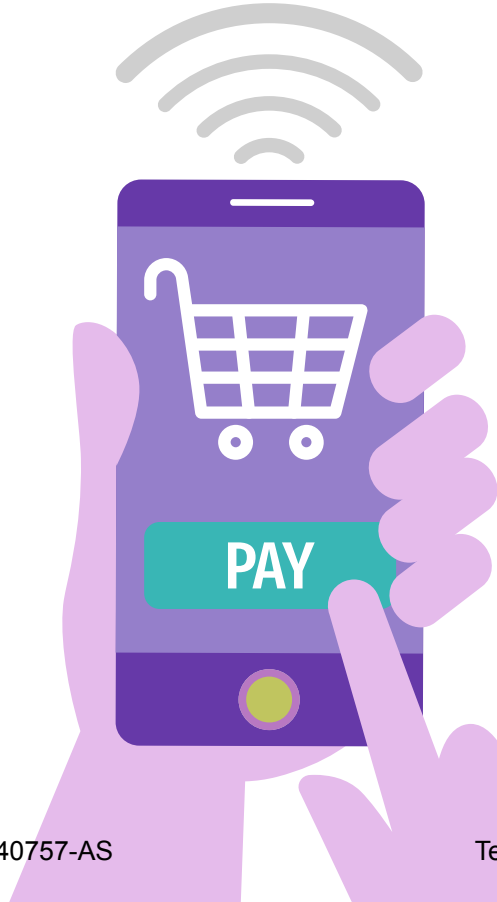


nopCommerce Observability



Index



Project Goal

nopCommerce with
Observability

Architecture Flow

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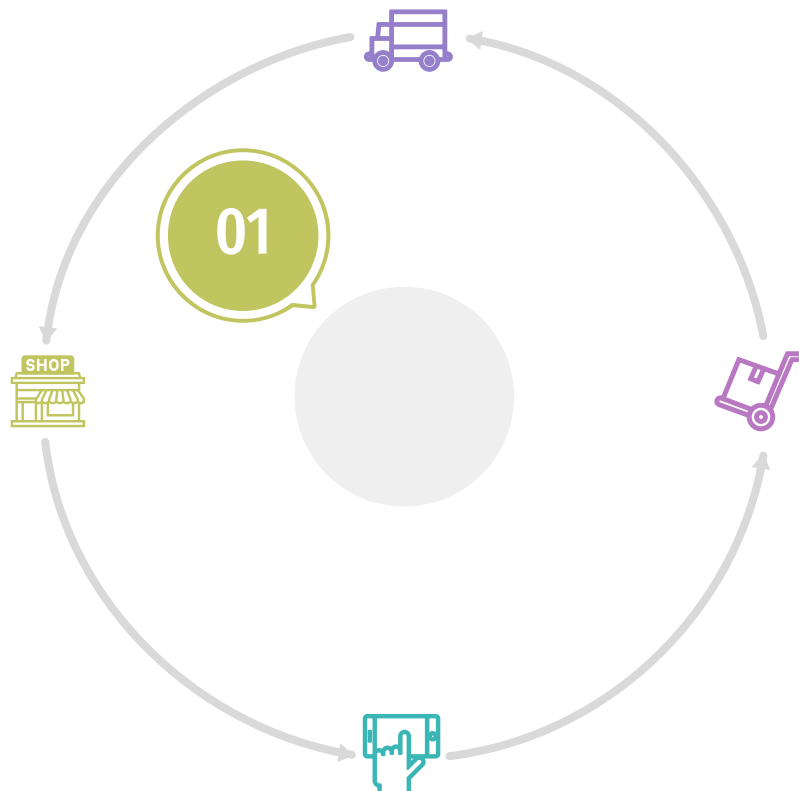
Tracers & Spans, Metrics and
Logs

Trade Offs, Demo, Conclusion

01

Project Goal

nopCommerce with
Observability



Goal

Spans & Tracers

Distributed tracing of the complete flow.

A

Metrics

Customized metrics

B

Logs

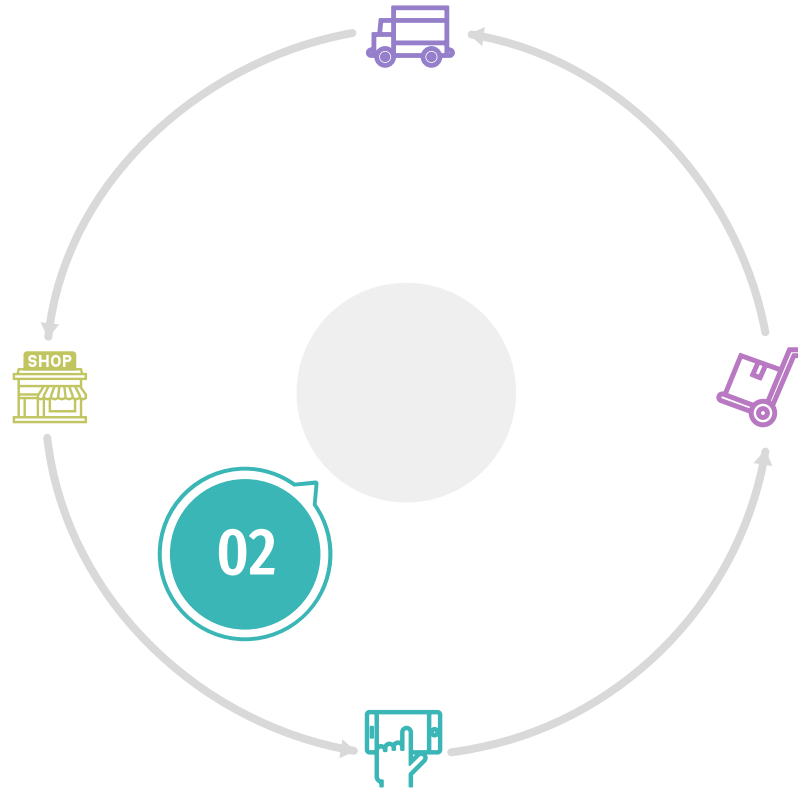
C



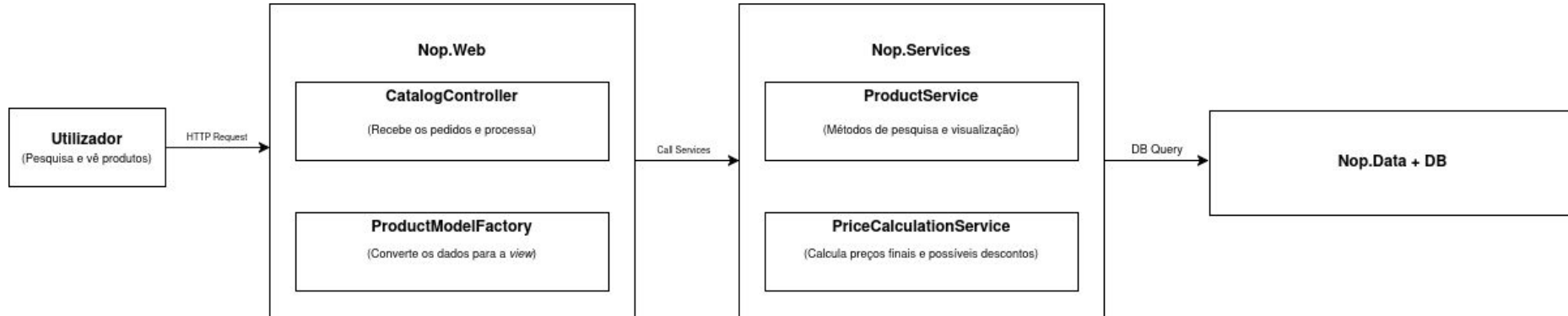
noCommerce with OpenTelemetry

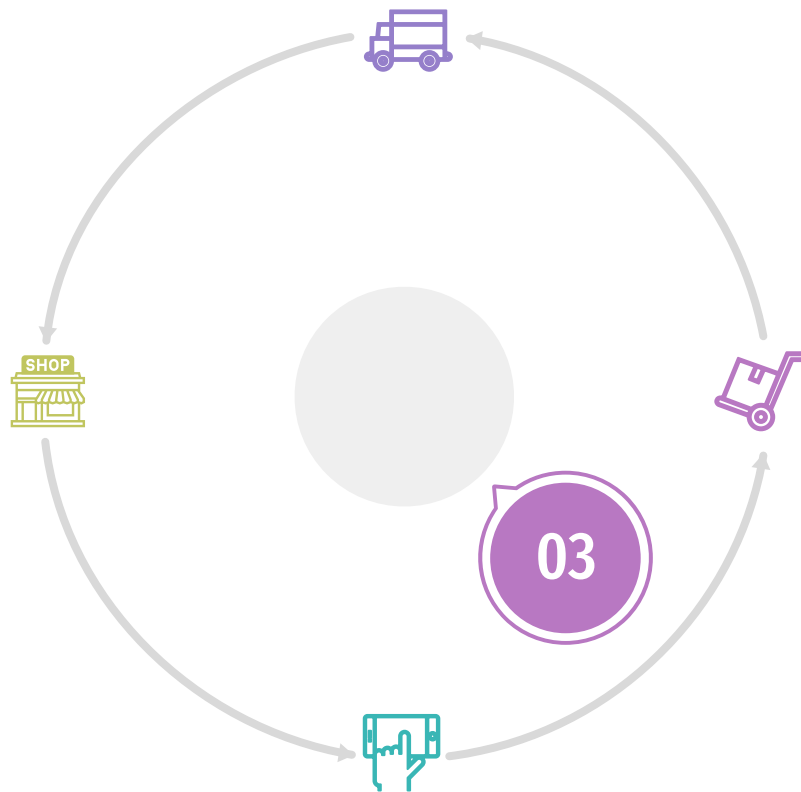
02 Architecture Flow

System Layers



Architecture





03 Instrumentation

Tracers & Spans, Metrics
and Logs

Instrumentation

Component	Spans	Metrics
CatalogController	Search, Autocomplete	-
ProductModelFactory	PrepareProductDetails	-
ProductService	SearchProducts, GetProductById, KeywordSearch	SearchDurationMs, SearchResultsCount, ProductDetailsLoadDurationMs
PriceCalculationService	GetFinalPrice, GetDiscountAmount	ProductPriceCalculationDurationMs



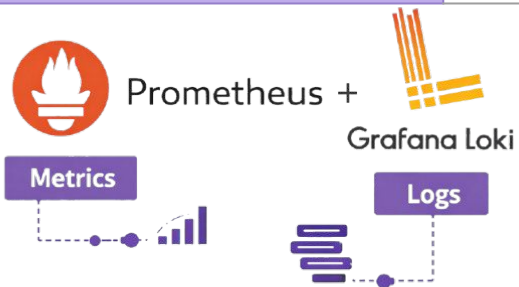
Spans & Tags

Spans	Tags
CatalogController.Search	'search.term', 'search.duration_ms', ...
Catalogue.SearchProducts	'catalog.has_keywords', 'catalog.result_count', ...
Catalogue.GetProductById	'catalog.product_id', 'catalog.product_found', ...
PriceCalculation.GetFinalPrice	' <u>product.id</u> ', 'discount.applied', 'price.final', ...



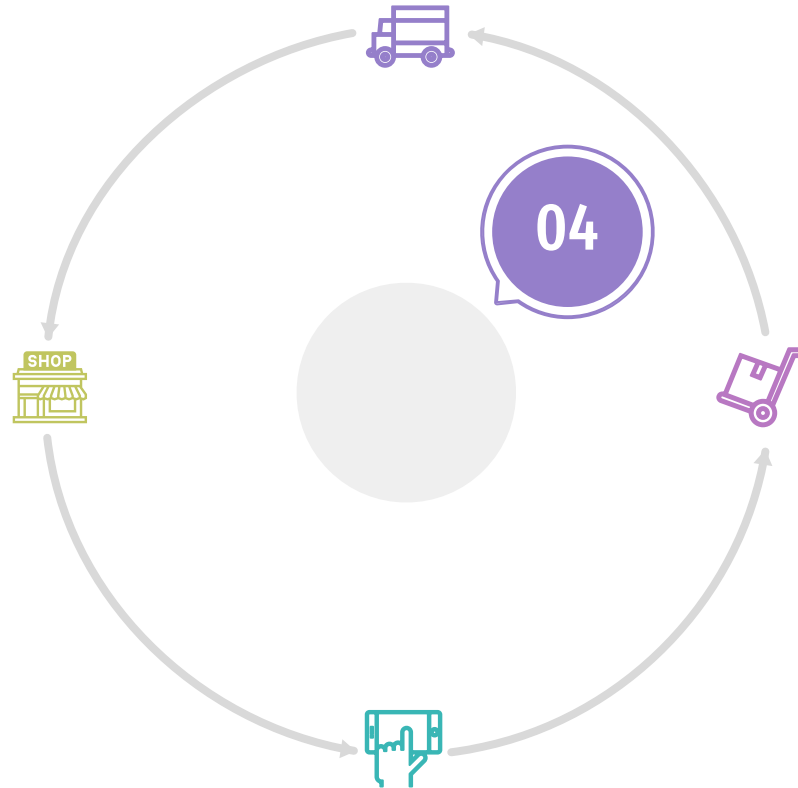
Metrics e Logs

Metrics	What is being measured?	What should I do if the value changes at 2 a.m.?
SearchDurationMS	Search latency	Check database and indexes
SearchResultsCount	Number of results per search	Check search index
ProductDetailsLoadDurationMs	Product loading time	Check cache and database queries
ProductPriceCalculationDurationMs	Price calculation time	Check the logic behind discounts and tier pricing



Observation - Logging is selective, therefore privacy is taken into consideration (only IDs, product names, etc.).

04 Trade Offs, Demo, Conclusion

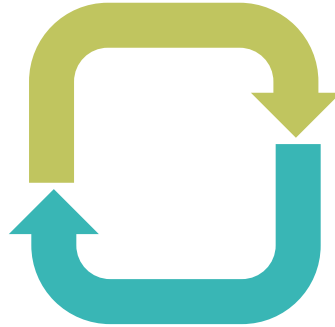


Trade Offs



What made it easier

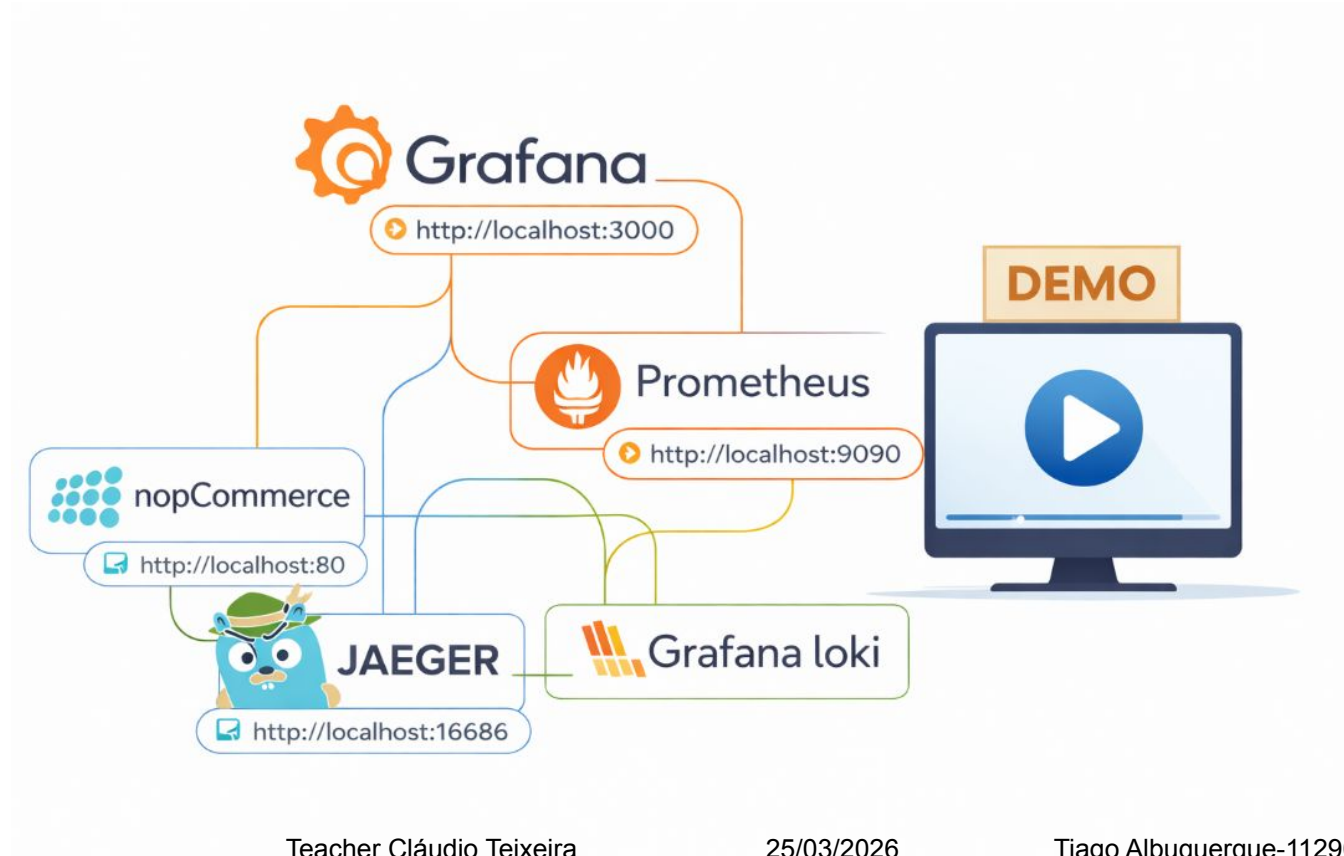
- Good code organization;
- Well-defined interfaces;
- Generalized dependency injection.

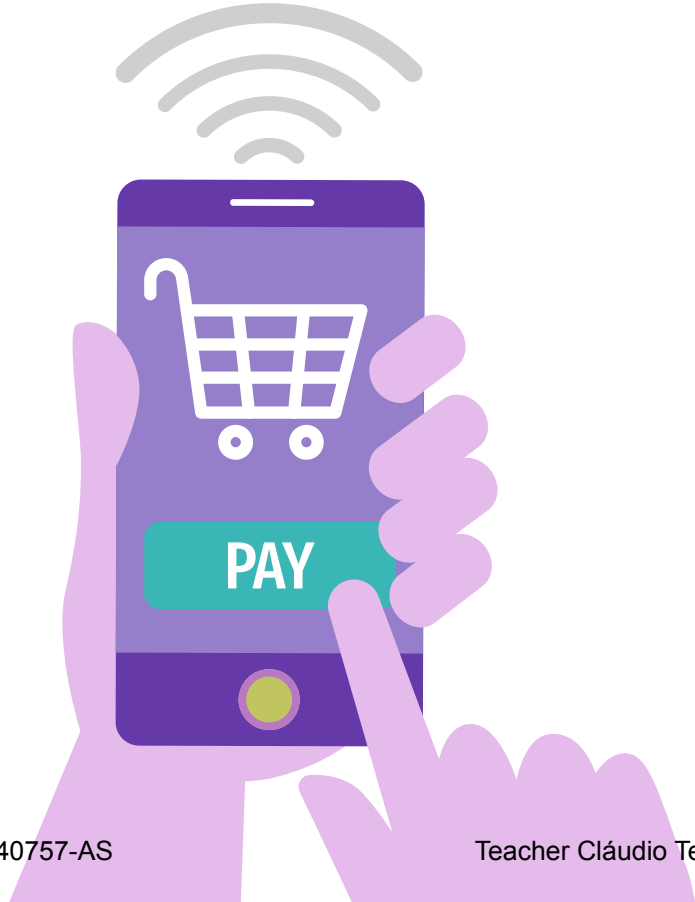


What made it difficult

- Dense methods;
- Cache mixed with method logic;
- Lack of a native ActivitySource.

Demo





Thanks!

Any Questions?